9.2 Inscribed Angles

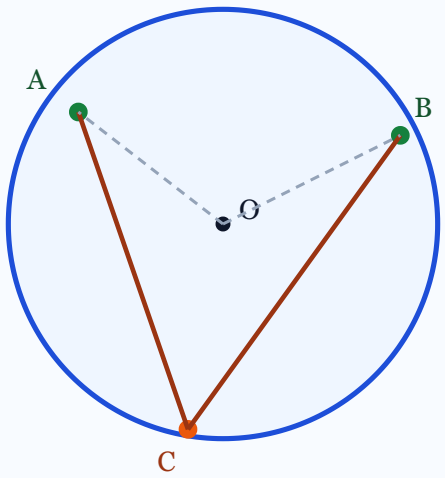
An **inscribed angle** has its vertex on the circle and its two sides are chords. The **Inscribed Angle Theorem** says an inscribed angle is always *half* its intercepted arc: $m\angle = \frac{1}{2} m\overset{\frown}{\text{arc}}$. Two immediate consequences: inscribed angles that intercept the *same* arc are congruent to each other, and an inscribed angle is half of the central angle on the same arc (since the central angle equals the arc). A special case called **Thales' Theorem** follows directly: if a triangle is inscribed in a circle with one side as a diameter, the angle opposite that diameter is always a right angle, because a diameter's arc is $180°$ and half of $180°$ is $90°$. This worksheet includes the reasoning behind the theorem, not just its use.

Examples

Example 1: Angle from an arc

An inscribed angle intercepts an arc of $84°$. Find the angle.

Solution



Inscribed angle C is half of arc AB.

Identify the theorem: an inscribed angle is always half of its intercepted arc, $m\angle = \frac{1}{2} m\overset{\frown}{\text{arc}}$.

Compute: $m\angle = \frac{1}{2}(84°) = 42°$.

Answer: $42°$.

Example 2: Arc from an angle

An inscribed angle measures $55°$. Find its intercepted arc.

Solution

Identify: this reverses the Inscribed Angle Theorem — since the angle is half the arc, the arc must be double the angle.

Compute: arc $= 2 \times 55° = 110°$.

Answer: $110°$.

Example 3: Angles on the same arc are congruent

Inscribed angles $\angle ACB$ and $\angle ADB$ both intercept arc AB: $m\angle ACB = (3x + 5)°$ and $m\angle ADB = (4x - 10)°$. Find x and the angle measure.

Solution

Angles intercepting the same arc are congruent:

$3x + 5 = 4x - 10$

$x = 15$.

Angle $= 3(15) + 5 = 50°$.

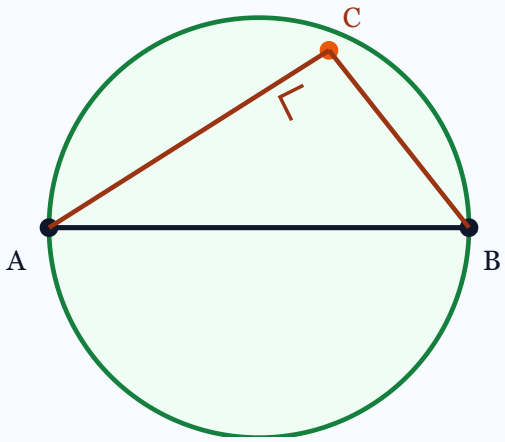
Answer: $x = 15$; each angle $= 50°$.

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Example 4: Thales' Theorem

Triangle ABC is inscribed in a circle with diameter $\overline{AB}$. $m\angle A = 35°$. Find $m\angle B$.

Solution



A triangle inscribed with a diameter as one side always has a right angle opposite that diameter.

Since $\overline{AB}$ is a diameter, its arc is a semicircle $(180°)$, so the inscribed angle at C is $\frac{1}{2}(180°) = 90°$.

The triangle's angles sum to $180°$: $m\angle B = 180° - 90° - 35° = 55°$.

Answer: $55°$.

Example 5: Central angle vs. inscribed angle

Central angle $\angle AOB = 100°$. Point C lies on the major arc. Find inscribed angle $\angle ACB$.

Solution

The central angle equals its intercepted arc: $m\widehat{AB} = 100°$.

The inscribed angle on that same arc is half: $m\angle ACB = \frac{1}{2}(100°) = 50°$.

Answer: $50°$.

Example 6: Angle from an arc

An inscribed angle intercepts an arc of $96°$. Find the angle.

Solution

Identify: apply the Inscribed Angle Theorem directly — the angle is half the arc.

Compute: $\frac{1}{2}(96°) = 48°$.

Answer: $48°$.

Example 7: Arc from an angle

An inscribed angle measures $63°$. Find its intercepted arc.

Solution

Identify: work backward from the angle to the arc by doubling.

Compute: $2(63°) = 126°$.

Answer: $126°$.

Example 8: Same-arc equation

Two inscribed angles on the same arc measure $(2x + 8)°$ and $(5x - 19)°$. Find x and the angle.

Solution

$2x + 8 = 5x - 19$

$27 = 3x \Rightarrow x = 9$.

Angle $= 2(9) + 8 = 26°$.

Answer: $x = 9$; angle $= 26°$.

Example 9: Semicircle triangle

A right triangle is inscribed with its hypotenuse as a diameter. One acute angle is $28°$. Find the other.

Solution

Identify: by Thales' Theorem, the angle opposite a diameter is always $90°$, so this is a right triangle with one leg-angle already known.

Compute the missing angle using the triangle angle sum: $180° - 90° - 28° = 62°$.

Answer: $62°$.

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Example 10: Why is the inscribed angle half the arc?

Prove the Inscribed Angle Theorem for the case where one side of the inscribed angle is a diameter: circle with center O, diameter $\overline{CB}$, and point A on the circle, forming inscribed angle $\angle ACB$.

Solution

Draw radius $\overline{OA}$. Since $OA = OC = r$, triangle OAC is isosceles, so its base angles are equal: $m\angle OCA = m\angle OAC = x$ (note $\angle OCA$ is the same angle as $\angle ACB$).

Because C, O, B lie on a straight diameter, $\angle AOB$ is the *exterior* angle of triangle OAC at vertex O.

By the Exterior Angle Theorem, an exterior angle equals the sum of the two remote interior angles: $m\angle AOB = m\angle OCA + m\angle OAC = x + x = 2x$.

But $\angle AOB$ is a central angle, so $m\angle AOB = m\overset{\frown}{AB}$. Therefore $m\overset{\frown}{AB} = 2x$, which means $x = \frac{1}{2}m\overset{\frown}{AB}$ — exactly the Inscribed Angle Theorem.

Example 11: Application — lighthouses on a bay

A curved shoreline is modeled as an arc of a circle. Two lighthouses sit on the shoreline, and a ship at a third point on the same circle sights both lighthouses. The arc between the lighthouses that does *not* contain the ship measures $130°$. Find the angle at which the ship views the two lighthouses.

Solution

The ship's viewing angle is an inscribed angle intercepting the arc between the lighthouses that does not contain the ship:

$m\angle = \frac{1}{2}(130°) = 65°$.

Answer: $65°$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Angle from arc

An inscribed angle intercepts an arc of $110°$. Find the angle.

Answer: **Verification:**

Problem 2: Arc from angle

An inscribed angle measures $37°$. Find its intercepted arc.

Answer: **Verification:**

Problem 3: Central vs inscribed

A central angle measures $150°$. Find the inscribed angle on the same arc.

Answer: **Verification:**

Problem 4: Same arc

Two inscribed angles on the same arc measure $(4x - 6)°$ and $(2x + 10)°$. Find x and the angle.

Answer: **Verification:**

Problem 5: Thales

Diameter $\overline{PQ}$ with R on the circle. $m\angle P = 42°$. Find $m\angle Q$.

Answer: **Verification:**

Problem 6: Angle from arc

Inscribed angle intercepts a $68°$ arc. Find the angle.

Answer: **Verification:**

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Problem 7: Arc from angle

Inscribed angle is $81°$. Find the intercepted arc.

Answer: **Verification:**

Problem 8: Half the central

Central angle is $88°$. Find the inscribed angle on the same arc.

Answer: **Verification:**

Problem 9: Thales

Inscribed triangle with diameter hypotenuse; acute angles $3x$ and $2x$. Find x and both angles.

Answer: **Verification:**

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Problem 10: Same arc

Angles $(x + 12)°$ and $(3x - 8)°$ intercept the same arc. Find x and the angle.

Answer: **Verification:**

Problem 11: Find angle

Arc measure $140°$. Find the inscribed angle intercepting it.

Answer: **Verification:**

Problem 12: Find arc

Inscribed angle $49°$. Find the arc.

Answer: **Verification:**

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Problem 13: Vocabulary

Where is the vertex of an inscribed angle located?

Answer: **Verification:**

Problem 14: Challenge

Two inscribed angles intercept arcs that together make up the whole circle, and one angle is 3 times the other. Find both inscribed angles.

Answer: **Verification:**

Problem 15: Challenge

A right triangle is inscribed with the hypotenuse as a diameter of length 26. One leg is 24. Find the other leg and both acute angles (nearest degree).

Answer: **Verification:**

✓ Answer Key

1. 55°
2. 74°
3. 75°
4. $x = 8$; angle 26°
5. 48°
6. 34°
7. 162°
8. 44°
9. $x = 18°$; angles 54°, 36°
10. $x = 10$; angle 22°
11. 70°
12. 98°
13. On the circle
14. 135° and 45°
15. Other leg = 10; angles ≈ 67° and ≈ 23°

Great work — keep practising!